ROBERT DELIA

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Professional Summary

Accomplished Computing & Information Technologies Specialist with an active DoD Security Clearance and a proven record of architecting mission-critical software solutions for the Air Force Research Lab (AFRL). Combines deep expertise in full-stack development, simulation infrastructure, and algorithmic design with a history of leading projects that automate complex defense workflows. Eager to apply advanced skills in software engineering and systems architecture to solve challenging problems in a high-impact technical role.

Relevant Experience

Air Force Research Lab (AFRL) | *Software Engineer / Research Assistant Intern* March 2025 – Present | Rome, NY

- **USAF Advanced Framework for Simulation, Integration, and Modeling (AFSIM) Automated Simulation & Code Generation:** Architected a Python translation engine for a DoD 6.1 foundational research initiative, automating the conversion of JSON mission schemas into the AFSIM simulation suites and reducing configuration time from hours to minutes.
- **Algorithmic Dependency Resolution:** Formulated dependency-resolution algorithms to automatically bind and synchronize prerequisite air combat mission assets (e.g., refueling tankers, Suppression of Enemy Air Defenses (SEAD)) across complex multi-aircraft timelines.
- **Master Air Attack Plan (MAAP) Automation & AI Planning:** Developed an interactive operational planning dashboard using Python and Plotly Dash to streamline scenario generation for a human-centered AI planning architecture on the $100M USAF Air Planning Environment for Execution Gaming Exploration Assessment and Refinement (APEX GEAR) program.
- **Technical Leadership & Collaboration:** Selected for mission-critical temporary duty (TDY) assignments and trainings to translate warfighter requirements into actionable software sprints; interfaced with simulation specialists, research scientists, and military Subject Matter Experts (SMEs) to align technical development with operational needs.

Utica Coffee Roasting Company | *Production Specialist & Barista* August 2021 – March 2025 | Utica, NY

- Maintained exceptional dependability and team communication in high-volume retail and warehouse environments.
- Safely operated and serviced industrial processing and automated packaging machinery.

Education

Rochester Institute of Technology (RIT) | *Bachelor of Science in Computing and Information Technologies* Expected Graduation: December 2026

- **GPA:** 3.85 / 4.00
- Honors: **Presidential Scholar**

Mohawk Valley Community College (MVCC) | *Associate of Science in Computer Applications Programming* Graduated: May 2023

- **GPA:** 4.00 / 4.00 (Graduated with Highest Honors)
- Honors: **Phi Theta Kappa Honor Society Member**

Engineering & Technical Projects

Makerspace Access Control & Web Infrastructure *Next.js, React, PostgreSQL, Raspberry Pi, Python*

- Co-engineered a full-stack facility management system with a 3-tier architecture, implementing secure session handling (RBAC), database-level constraints, and an edge-computing verification unit (Raspberry Pi, RFID) for sub-second member authentication.

AFSIM Automated Scenario Generation Engine *Python, JSON Schema, REST APIs, Algorithmic Logic*

- Architected an end-to-end Python script-generation engine that ingests complex operational air combat plans (Specifically: Air Operations Center (AOC) MAAP) JSON plans and automatically compiles fully executable simulation suites for AFSIM, featuring dynamic flight-dependency resolution and persistent data store integration.

Operational Scenario Authoring & Planning Dashboard *Python, Plotly Dash, REST APIs, Geospatial Mapping*

- Developed an interactive dashboard for the APEX GEAR program using Plotly Dash, implementing a custom node-graph interface to trace mission assets and embedding geospatial map overlays and schema validation pipelines to ensure data integrity.

Airline Enterprise Relational Database System *MySQL, Relational Schema Design, ERD/DFD Modeling*

- Architected and normalized a comprehensive relational database schema for a simulated airline enterprise, modeling complex operations (scheduling, booking, maintenance) and formulating optimized SQL queries for business intelligence reporting.

Key Skills

Programming Languages: Python, C++, Java, JavaScript, TypeScript, SQL, Bash

Databases & Cloud: PostgreSQL, Docker, Git, Linux, Vercel, Railway

Systems & Hardware: Raspberry Pi, RFID Integration, Hardware-Software Interfacing

Web & Frameworks: React, Next.js, Node.js, RESTful APIs, HTML5/CSS3

Modeling, Simulation & AI: AFSIM, Agentic AI Tooling, Generative AI Systems

Engineering Methodologies: Agile/Scrum, Full-Stack Architecture, Cross-Functional Collaboration